

Akhil Prasad Chinthala

AI / ML Engineer | LLM & Computer Vision Systems

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SUMMARY

Final-year IT student (CGPA 8.64) shipping LLM-powered features as an AI Engineering Intern at AMIK Technologies and authoring agentic coding benchmarks for frontier-model evaluation as a freelance AI trainer at Handshake AI. Hands-on across Python backend and applied AI: FastAPI model-serving endpoints, retrieval-augmented generation and LLM evaluation, and real-time computer vision (CNN emotion recognition at 70.6% on FER-2013, about 8 ms per frame on CPU). Seeking AI/ML or Python backend engineering roles building production LLM and vision systems.

EXPERIENCE

AI Engineering Intern

Apr 2026 – Present

AMIK Technologies

Hyderabad, India

- Build production retrieval-augmented generation (RAG) pipelines over internal documents — chunking, embedding generation, and vector similarity search — replacing manual lookup with grounded, citation-backed answers.
- Design and version prompt templates with structured outputs and function calling, standardising LLM responses into schema-validated JSON consumed directly by downstream services.
- Ship an automated evaluation harness scoring retrieval relevance, answer faithfulness, and latency across model and prompt versions — turning subjective review into regression-tested releases.
- Serve models behind FastAPI endpoints with token streaming, per-request cost logging, and provider fallback for resilience under rate limits.
- Reduce inference cost and latency through chunk-size tuning, semantic caching, and context compression; integrate features into the live product with the backend team.

Freelance AI Trainer — Agentic Benchmark Authoring

Jul 2026 – Present

Handshake AI

Remote

- Author terminal-based benchmark tasks used to evaluate frontier AI coding agents — 102 tasks delivered across 10 domains, including systems infrastructure, model-training and ML pipelines, data processing/ETL, databases, and interactive simulation.
- Ship each task as a reproducible package — natural-language specification, Dockerised environment, automated test-based verifier, and a working reference solution — built to clear automated QA gates and human review.

Software Development Intern — Python

Oct 2025 – Nov 2025

Oasis Infobyte

Remote

- Delivered five Python applications on a structured internship track, each specified, built, and demoed independently.
- Highlights: a speech-recognition voice assistant that parses spoken intent into system commands, a REST-integrated weather CLI, and a multi-client TCP chat server handling concurrent sessions.

PROJECTS

RBI FEMA RAG — Evaluated Regulatory Q&A | *Python, sentence-transformers, Ollama*

2026

- Built a fully local RAG system over RBI's foreign exchange Master Directions (~570 pages): hybrid BM25 + dense retrieval, cross-encoder reranking and cited answers, evaluated against 186 of RBI's own published FAQ question-answer pairs.
- Replaced an unreliable LLM relevance judge (precision 0.12–0.15) with a calibrated embedding judge (95% agreement on checked labels) to build pooled relevance labels; reported bootstrap confidence intervals and paired significance tests. Best retrieval: **83% hit@5 vs 59% for BM25**.
- Found the LLM answered 67% of questions the documents could not answer; added a cross-validated retrieval-confidence gate that raised **correct refusals from 33% to 75%**.

CAG Real-Time Emotion Tracker | *PyTorch, OpenCV, scikit-learn, Streamlit, Python*

2024 – 2026

- Trained a 5M-parameter CNN with squeeze-and-excitation attention on FER-2013 to **70.6% test accuracy across 7 emotion classes** (human agreement $\approx$ 65%), up from 38% for the team's hand-crafted-feature baseline.
- Replaced the classifier head with a cache-augmented lookup — one matrix multiply against in-memory per-class prototype embeddings — holding the full webcam pipeline at **about 8 ms per frame on CPU**.
- Led a 3-person team from architecture through delivery; shipped a Streamlit dashboard and presented results to a faculty review panel.

- E2EE Chat — End-to-End Encrypted Group Chat** | *Python, cryptography, PyQt5, Sockets* 2026
- Rebuilt my internship chat app with end-to-end encryption: on-device X25519/Ed25519 identities, AES-256-GCM messages with per-recipient key wrapping under ephemeral X25519 keys, and Ed25519-signed envelopes; the relay server only stores ciphertext it cannot read.
 - Defended against a malicious server with key pinning and Signal-style safety numbers; **14 tests** show tampering, forged senders and a **server key-swap attack** are rejected; also fixed an HTML-injection bug in the original GUI.
- Full-Stack Booking Platform with Voice AI Assistant** | *React, Vite, Node.js, Express, Web Speech API* 2025
- Built an end-to-end web application — React/Vite frontend backed by an Express REST API — deployed live in production for a real business client (example: a dental clinic), handling genuine customer bookings.
 - Integrated a voice-enabled AI assistant (*DentalBot*) that answers the client’s service FAQs and converts visitors into bookings.
 - Automated the full booking workflow: appointment data through the REST API, Google Sheets sync for the client’s staff, and automated email confirmations.
- Hand-Gesture Media Controller** | *MediaPipe, OpenCV, Python, AppleScript* 2024
- Mapped MediaPipe hand-landmark detection to macOS volume and playback controls for fully touchless media operation.
 - Sustained **30 FPS with sub-20 ms gesture-response latency** by optimising the landmark-to-action inference loop.
- Developer Portfolio** — akhil-prasad09.github.io | *Next.js, React, TypeScript* 2026
- Built a single-page developer portfolio with animated project, benchmark, and experience showcases, exported as a static Next.js site and deployed to GitHub Pages.
- Green Basket — E-Commerce Web App** | *React, JavaScript, CSS3* 2024
- Responsive grocery storefront built as a React app, with dynamic product filtering, cart state management, and a full checkout flow.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, C, C++

LLM & Generative AI: Retrieval-Augmented Generation (RAG), prompt engineering, embeddings & vector search, function calling / structured outputs, LLM evaluation, semantic caching

Machine Learning & CV: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, MediaPipe, NumPy, Pandas, fine-tuning

Deployment & Tooling: FastAPI, REST APIs, Git & GitHub, Linux (Ubuntu), model serving, latency & cost profiling

Web & Data: React, Vite, Node.js, Express, MySQL, MongoDB, vector databases

Foundations: Deep Learning, Computer Vision, NLP, Data Structures & Algorithms, Operating Systems, Networks

Spoken languages: Telugu (Native), English (Fluent), Hindi (Conversational)

EDUCATION

Matrusri Engineering College — B.Tech, Information Technology — CGPA 8.64	2023 – 2027
Sri Chaitanya Junior Kalasala — Higher Secondary, Science & Mathematics — 93%	2023
The Hyderabad Public School — Secondary School — 89%	2021